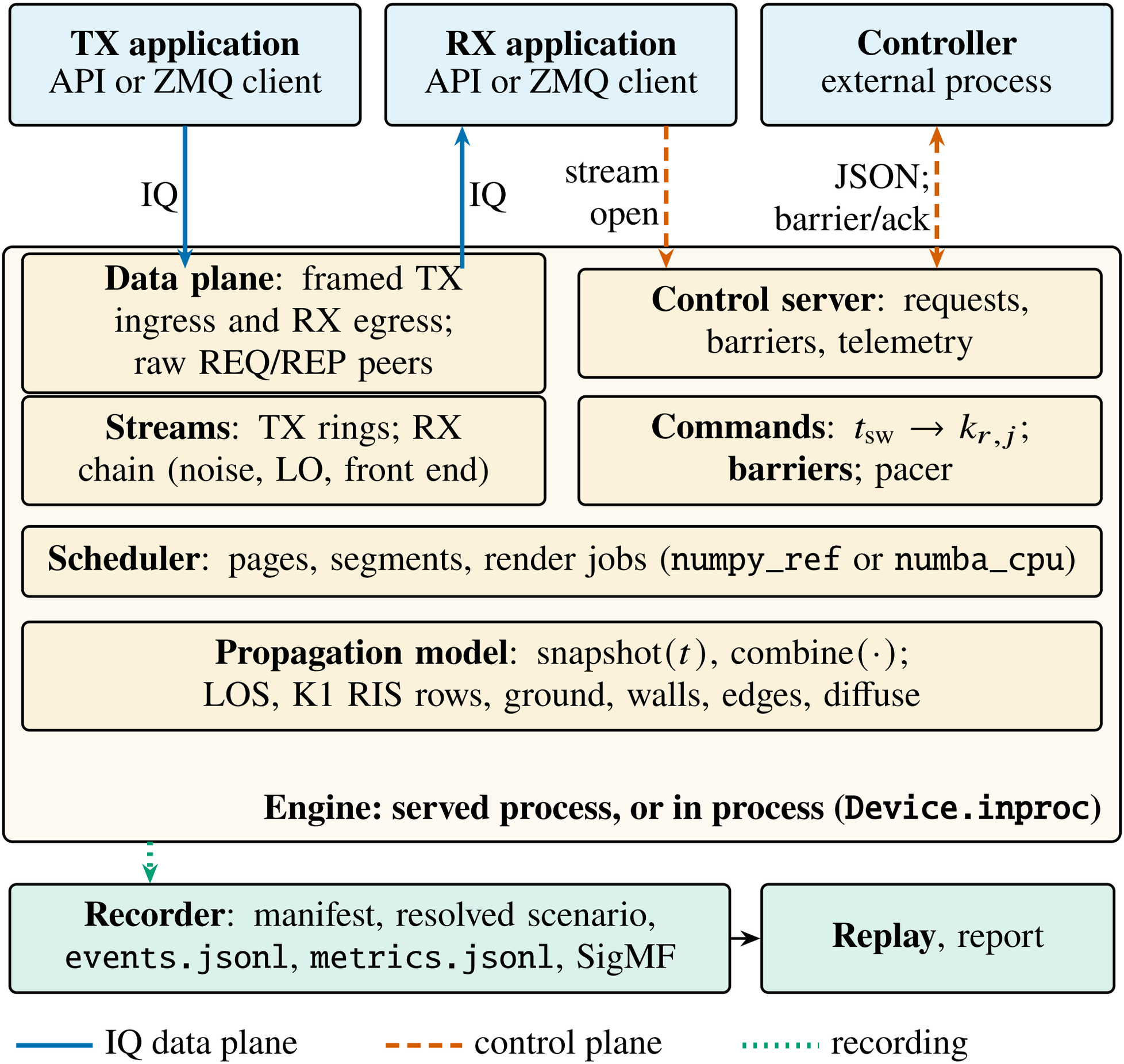

TX application
API or ZMQ client
RX application
API or ZMQ client
Controller
external process
IQ
IQ
stream open
JSON; barrier/ack
Data plane: framed TX ingress and RX egress; raw REQ/REP peers
Control server: requests, barriers, telemetry
Streams: TX rings; RX chain (noise, LO, front end)
Commands: $t_{sw} \rightarrow k_{r,j}$; **barriers**; pacer
Scheduler: pages, segments, render jobs (`numpy_ref` or `numba_cpu`)
Propagation model: snapshot(t), combine($\cdot$); LOS, K1 RIS rows, ground, walls, edges, diffuse
Engine: served process, or in process (`Device.inproc`)
Recorder: manifest, resolved scenario, `events.jsonl`, `metrics.jsonl`, SigMF
Replay, report
IQ data plane
control plane
recording